



Interaction of Enterprise Resource Planning and Sustainability in Digital Transformation Environment: A Bibliometric Analysis



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Abstract: Sustainability has become a pivotal concern for organizations and politicians worldwide, leading to accelerating focus on Enterprise Resource Planning (ERP) within this context. This study aims to delineate a structural network map of the fundamental relationship between the concepts of ERP and sustainability, their evolutionary dynamics, and future research trends as revealed in the international literature. To this end, articles relevant to the subjects were downloaded from the Web of Science (WoS) Core Collection database. After meticulous data cleaning, a bibliometric analysis of 2,035 articles was performed using the VOSviewer software. Keyword network analysis revealed 122 unique links, a Total Link Strength (TLS) of 2,485, and 9 distinct thematic clusters, indicating a highly integrated topology. The examination of the network map empirically showed that the concept of sustainability was the largest cluster, followed by the ERP system. Besides, the study discovered that the concepts of Corporate Social Responsibility (CSR), Stakeholder Engagement, and Environmental Policy were the most prominent keywords. Taking into account of geographical locations, the USA ranked first with 16,439 citations. Trend studies revealed a shift towards high-technology focused research in areas such as AI, cloud-based ERP systems, information systems, circular economy, and strategic decision making. Based on these findings, the study offered researchers a roadmap to access resources more easily and effectively so as to further their ERP and sustainability studies in the digital environment.

Keywords: Enterprise Resource Planning; Sustainability; ESG; Digital transformation; Corporate Social Responsibility; AI

1. Introduction

The trend of rapid changes in the business world are propelling organizations to adopt Enterprise Resource Planning (ERP) systems to balance their sustainability and corporate efficiency goals (Barna, 2024). The integration of sustainability and ERP concepts is a significant factor in improving social, environmental, and economic performance reports, and has become a fundamental process in adopting a sustainable management approach (Alzahmi et al., 2025). A business could improve its sustainability performance if it meets its current needs by utilizing environmentally-friendly technology, if the products and services it offers reduce their ecological footprint, and if it limits products and services that substantially harm the environment (Ekins, 1999). Assume a business fails to integrate its corporate data with its sustainability goals, it may face difficulties in process management (Alzahmi et al., 2024). Studies in the literature examining the intersection of ERP systems and sustainability strongly emphasized the integration of resource consumption management and environmental footprint tracking into information systems (Ebirim et al., 2024). Supporting ERP systems with real-time data analytics helps businesses comply with environmental regulations and, in the long term, positions itself as a strategic strength in terms of corporate resilience and operational efficiency (Pugna & Boldeanu, 2025). Recent studies in the literature demonstrated that the relationship between enterprise resource management and sustainability was strengthening, and there was an essential trend towards scientific network mapping studies

(Praharaj & Chhatoi, 2025). In this connection, the growth in publications examining the role of ERP systems in sustainable business models demands a comprehensive mapping of these concepts (Koçyiğit et al., 2023; Özarı & Can, 2023). According to the findings in the ERP literature, researchers mostly focused on strategic management and implementation, while studies on sustainability are mostly recurring Corporate Social Responsibility (CSR) and green computing (Barna, 2024).

Hasan et al. (2017) proposed several performance alternatives for incorporating sustainability elements into the ERP system. These options, encompassing economic, social, and environmental dimensions, offer a fundamental framework for Sustainable ERP (S-ERP). In this study, 63 statements of sustainability were identified, of which 21 were economic, 16 social, and 26 environmental. Pérez Estébanez (2024) conducted a survey to measure the satisfaction levels of company officials with S-ERP systems. The results indicated that the type of module included in the system had a positive and crucial effect on satisfaction with S-ERP. Furthermore, it was found that an expansion in the number of accounting modules led to increased complexity in the system, requiring more effort in using new technologies effectively.

Based on existing studies in the literature, it has been observed that the concepts of ERP and sustainability were addressed in different disciplines such as operational management, information systems, business ethics, and environmental transformation. The failure to combine synonymous terms such as ERP, and ERP systems could obscure the center of gravity of some trends. The crises caused by global climate change and the normative forces in international sanctions have transformed sustainability from an element of corporate reputation into an operational necessity. Currently, companies are facing stakeholder pressure to not only declare their Environmental, Social, and Governance (ESG) performance but also to make this performance transparent, verifiable, and reportable with real-time data. Therefore, the necessity of a socio-technical infrastructure that will connect macro-level sustainability goals with micro-level corporate operations is coming to the forefront. In this context, ERP systems, which provide data flow for companies, are a key element in integrating sustainability inputs with operational processes. This study aims to present from a holistic perspective scientific network maps of isolated ERP systems and sustainability concepts across different disciplines, without methodological noise and conceptual fragmentation. The main theme of this study is to trace the trends in the ERP and sustainability literature between 2010 and 2025, identify prominent actors (authors, institutions, and countries), and explore areas of trend research. In this context, the study systematically mapped the literature through author and co-author analyses, citation network analysis, and keyword bibliographic matching analysis, to offer original contributions intended to direct future research. To this end, 2,035 articles published between 2010 and 2025 and obtained from the Web of Science (WoS) Core Collection database after rigorous data cleaning, were subjected to bibliometric analysis using VOSviewer software.

2. Literature Review

ERP systems are indispensable to managing corporate assets with optimum efficiency and optimizing operational processes (Agarwal & Gupta, 2024; Odenwald & Berg, 2014). It enables companies to reduce costs, integrate resources, exchange information, and manage operational processes more effectively, efficiently, and smoothly (Lotfy, 2015). An ERP system, which facilitates the integration of all units in business processes, offers businesses the ability to monitor changes in performance in real time, while also supporting the decision-making process by providing rapid analysis capabilities (Lin et al., 2006). Companies adopting this system develop sustainability strategies that increase efficiency in operational processes while reducing their environmental footprints. The concept of sustainability is about meeting the needs of the present without jeopardizing the needs of future generations. Practices in corporate sustainability significantly contribute to reducing ecological impacts, strengthening relationships with stakeholders, and increasing profitability. Businesses demand a system that integrates operational processes, synchronizes sustainable management for specific purposes, and allows centralized database management and resource and status tracking Barna (2024). Within the scope of ERP, sustainability aims to provide a holistic perspective and reporting capabilities by centrally managing corporate processes, human resources, and the product lifecycle from a single location (Chofreh et al., 2020). ESG criteria are of critical importance today for investors, customers, employees, and regulatory bodies. These stakeholders thoroughly analyze the environmental and social dimensions of companies' operational processes and practices (Clark et al., 2015). The United Nations Sustainable Development Goals, adopted in 2015 with the consensus of 193 countries, provide a fundamental framework for overcoming urgent global challenges and creating a sustainable development model. Focusing on goals such as eradicating extreme poverty, reducing inequalities, and protecting the planet by 2030, the United Nations 2030 Sustainable Development Agenda sets out 17 key goals navigating a wide spectrum of challenges from poverty to climate change and environmental degradation (United Nations, 2015).

Businesses struggling to achieve their corporate sustainability goals tend to understand more about the critical importance of integrating digital transformation with sustainability practices. This approach, described in the literature as “Dual Transformation”, is based on the synchronized implementation of digitalization and

sustainability strategies. This process renders it possible for technology not only to optimize operational efficiency but also to integrate environmental and social responsibility into long-term strategies (Christmann et al., 2024).

The role played by S-ERP is crucial in implementing sustainability strategies at the operational level. These systems integrate the basic dimensions of sustainability as follows:

1. **Social Sustainability:** S-ERP leverages social data to enrich decision-making mechanisms, incorporating principles of social equity and environmental responsibility into business processes. Minimizing risks arising from volatile economic dynamics, this system supports organizations in achieving their long-term social sustainability goals. Essentially, S-ERP offers a comprehensive structure focused on managing the principles of “people, planet, and profit”.

2. **Financial Sustainability:** S-ERP structures strategic decision-making processes with financial performance indicators. By adopting a proactive risk management approach against economic fluctuations, it aims to secure the long-term financial stability and resilience of businesses.

3. **Environmental Sustainability:** S-ERP optimizes decision-making processes by analyzing ecological data. This approach devises a systematic framework that aims to create value for both the business and society while effectively managing environmental risks.

4. **Corporate Sustainability:** S-ERP strengthens corporate governance by processing organizational data with a strategic approach. This system contributes to the creation of a sustainable business model that adds value for stakeholders by managing corporate risks (Anjaria, 2025).

CSR is defined as a self-regulated business model that incentivizes businesses to be accountable to their stakeholders and the public (Fatima & Elbanna, 2023). CSR initiatives are generally classified into four main categories: environmental, ethical, philanthropic, and economic. Each of these dimensions makes a holistic contribution to the company’s overall CSR strategy. For example, environmental responsibility supports sustainability in businesses through environmentally-friendly practices such as reducing energy consumption, using renewable energy resources, and waste management (Tilt, 2016). S-ERP systems refer to structures of ERP that prioritize sustainability principles at all stages of the value chain (Anaya et al., 2023).

Meanwhile, Bhagat (2025) defined ERP systems as a structure that strengthens transparency and accountability through explainable AI, robust governance frameworks, and real-time data processing capabilities compatible with the governance and social dimensions of ESG. The author argued that even if these systems did not directly focus on the ESG concept or the digital transformation of the private sector, they functioned as a strategically responsive and transformation-oriented tool in public sector operations, addressing social goals. Cloud-based solutions contribute to more effective system management by accelerating information access and increasing energy efficiency. Emphasizing reducing carbon footprint and optimizing operating expenses, cloud computing also lays a strategic foundation for electronic waste management, recycling processes, and reuse strategies. Furthermore, data collection and analysis processes carry out more efficient waste management through digital platforms and assist in the monitoring of product performance (Yüzgenç & Aydemir, 2023).

AI-integrated ERP solutions could minimize the risk of “greenwashing” by detecting discrepancies in sustainability data. These technologies optimize carbon emission calculations, water consumption monitoring, and waste management processes, thereby strengthening the transparency and auditability of ESG reporting (Pugna & Boldeanu, 2025). Wang et al. (2025), in their empirical research on Chinese firms, demonstrated that AI integration in ERP systems significantly elevated ESG performance via supporting green innovation.

3. Methodology

This study adopted a quantitative research approach using bibliometric analysis and scientific mapping to reveal the intellectual structure, conceptual network topology, and temporal evolution of the global literature in the field of ERP and enterprise sustainability and ESG integration. Bibliometric analysis is a frequently used and effective research method for systematically and rigorously examining extensive scientific literature (Donthu et al., 2021). In this study, VOSviewer (VOS-Visualization of Similarities) (Version 1.6.20) software was employed because it could easily visualize and interpret large bibliometric maps in advanced visualization and distance-based network mapping algorithms (Van Eck & Waltman, 2010).

To ensure high indexing reliability and international reputation standards for the analysis data, the WoS Core Collection, including Science Citation Index Expanded, Social Sciences Citation Index, and Arts & Humanities Citation Index, was selected as data source owing to its prestigious status as a database with global academic output. On November 26, 2025, a search of the WoS database for the keywords “ERP” and “sustainability” across all fields yielded 4,665 publications. The document types were selected as “Article” and “Review Article”, and English was chosen as the reporting language. Related categories included business, management, information systems, industrial engineering, environmental science, and green sustainability. Studies on medicine, health, psychology, and natural sciences were excluded. To capture all potentially relevant studies in the literature, Boolean logic was used in the search query: ("Enterprise Resource Planning" OR "ERP" OR "ERP systems") AND ("sustainability" OR "sustainable" OR "Corporate Social Responsibility" OR "CSR" OR "ESG" OR

"environmental policy"). As a result of this filtering, a raw database (full records and cited references) consisting of a total of 2,035 articles from 2010–2025 was downloaded in txt format for analysis. In bibliometric network analyses, directly loading raw data into the software leads to semantic clutter and consequently “methodological noise”, due to different spelling variations of the same concept (e.g., ERP, ERP systems, and Enterprise Resource Planning) or abbreviations (Landauer et al., 1998). To prevent this methodological deviation and maximize the validity and reliability of the analysis, a rigorous data cleaning and thesaurus/alias file was created before the analysis. If the data is loaded directly into the program without cleaning, words with the same meaning but different spellings will appear as separate bubbles in the keyword map, hence mitigating the quality of the analysis. Therefore, a “thesaurus.txt” file was created and integrated into the program before commencing the analysis. In the thesaurus.txt file, two columns, “label” and “replace by” were created to combine terms that express the same concept but are written in different ways. For example, variations such as “enterprise resource planning”, “ERP systems”, and “enterprise systems” were combined under the “ERP” bubble; the terms “sustainable development” and “sustainable” were combined under the “sustainability” bubble; “environmental social and governance” was combined as “ESG”, and “sustainable ERP” was combined as “green ERP”. The downloaded dataset was then transferred in parts to the VOSviewer program and analysis began on the same date. The following analyses were performed on the cleaned dataset to decipher different layers of the literature: Co-authorship analysis, author citation analysis, country citation analysis, keyword co-occurrence analysis, organization citation analysis, co-citation analysis of co-authors, and bibliographic coupling analysis of documents.

4. Data Analysis

4.1 Co-Authorship Analysis

Co-authorship analysis was performed using VOSviewer to illuminate the collaboration networks among authors. In the analysis, the author was selected as the unit when the full counting method was applied and the association strength algorithm was preferred for normalization. The minimum 1 publication and minimum 1 citation criterion (threshold values) were taken as the basis. The results of the analysis are given in Figure 1.

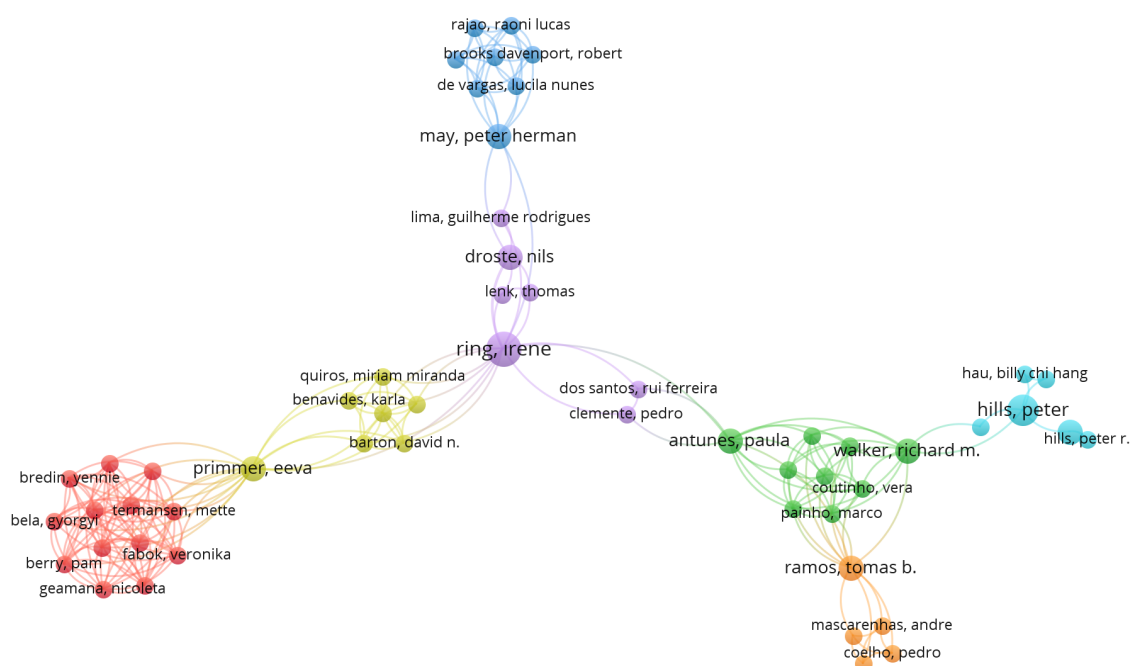


Figure 1. Collaboration and co-authorship relationships among authors

A co-authorship of authors analysis, conducted to uncover the intellectual collaboration structures shaping the ERP and sustainability/ESG literature, identified 55 key authors (items) meeting a defined threshold. The network structure mapping the academic collaboration relationships among these researchers was characterized by a Total Link Strength (TLS) of 228.

In the co-authorship analysis conducted on an author-by-author basis, a ranking was made using the TLS criterion to determine the researchers' impact on the literature and their central positions in collaborative networks. Table 1 shows the background network matrix ranked in descending order according to TLS values. Hoffman,

David J. (TLS = 40), Jiang, Chong (TLS = 37), Stebbins, Katherine (TLS = 34), Van Erp, Tim (TLS = 33), and Heinz, Gary H. (TLS = 30) were identified as the most central and founding actors in the field. These researchers were found to be central actors in the 2,035 articles analyzed, both in terms of publication volume and high total connectivity strength, assuming intellectual leadership within their own clusters. These authors were identified as the most influential figures dominating the ERP and sustainability literature, thanks to both their high publication volumes and their strategic bridging positions within the network. The results showed that, as frequently emphasized in the bibliometric literature, the most cited authors were not always the most connected authors. In other words, researchers whose theoretical contributions stand out with high citation numbers could differ from those who occupy central positions in productivity and collaboration networks (Aksay & Ağtaş, 2025).

Table 1. Prominent authors in co-authorship analysis

Author	No. of Documents	No. of Citations	TLS
Hoffman, David J.	10	193	40
Jiang, Chong	6	235	37
Stebbins, Katherine	9	186	34
Van Erp, Tim	7	273	33
Heinz, Gary H.	8	142	30

Note: TLS = Total Link Strength.

4.2 Author Citation Analysis

To perform an “Author-Based Citation Analysis” targeted at determining the scientific impact levels and institutional reference sources in the literature, output files from WoS were exported as plain text, including Full Record and Cited References, and then uploaded to VOSviewer. The following commands were used for analysis: Create map → Create a map based on bibliographic data → WoS files → Type: Authors → Analysis: Citation. The counting method was selected as full counting, and the normalization as association strength; VOS defaults were used for placement and clustering. In the author citation analysis, a threshold value of at least one publication and at least one citation was entered as the criterion, and then a network map was created to articulate the citation networks. The resulting network map is given in Figure 2.



Figure 2. Authors' citation network analysis

An analysis of 2,813 interconnected units revealed a total of 40 clusters, 22,874 connections, and a total connection strength of 24,806. Clustering becomes more pronounced as the number of connections increases. In the visual representation, each point (node) represents an author, the size of the points indicates the number of citations received by the author, and the colors represent clusters. The lines (links) between authors indicate citation or co-citation relationships.

In the author citation analysis performed, the “intellectual capacity of the authors” (individual authors) was used as the unit of analysis instead of individual articles (documents). Due to the data aggregation algorithm of the VOSviewer software, all works published by an author in different years and the total citations obtained from these works were cumulatively consolidated into a single node. Therefore, the absence of a specific publication year in parentheses next to the author’s name in the network map is a methodological necessity stemming from the macro-level author-based data integration and reflects the author’s cumulative impact on the relevant literature.

As a result of the “Author-Based Citation Analysis” conducted to determine the scientific impact levels and corporate reference sources in the literature concerning ERP and sustainability/ESG integration, the top 5 authors with the highest number of citations were identified. According to the sorting matrix created in Excel from largest to smallest, the first three authors: Schaltegger, Stefan (1,962), Wagner, Marcus (1,751), and Dangelico, Rosa Maria (1,642), stood out as the most dominant figures forming the intellectual backbone of the field with their high number of citations. These authors were followed by Kolk, Ans (1,204) and Baumgartner, Rupert J. (1,198) in terms of their citations.

As shown in Table 2, an examination of the overall topology of the network structure reveals that the top five authors were not concentrated in a single central cluster on the map but were instead distributed across parallel, linear clusters. This parallel distribution proved that these leading researchers, who received the most citations, tackled the issues of ERP and sustainability from different disciplinary perspectives (e.g., green supply chain,

information system management, or corporate sustainability theories). Consequently, the works of these 5 most cited actors should be considered as fundamental reference points that determined the course of the development in the literature.

Table 2. Top 5 most cited authors in author citation analysis

Author	No. of Documents	No. of Citations	TLS
Schaltegger, Stefan	7	1,962	241
Wagner, Marcus	4	1,751	140
Dangelico, Rosa Maria	5	1,642	135
Kolk, Ans	8	1,204	200
Baumgartner, Rupert J.	6	1,198	150

Note: TLS = Total Link Strength.

4.3 Citation Analysis of Countries

A “Citation Analysis of Countries” study was conducted to determine the global impact, prestige, and geographical centers of gravity of scientific output in the ERP and sustainability/ESG literature. This analysis revealed not only the capacity of countries in quantitative publication production but also the global impact and quality of the academic knowledge they produced.

WoS output was exported as plain text in the form of Full Record and Cited References to the VOSviewer program. For analysis, the following steps were used: Create map → Create a map based on the bibliographic data → WoS files → Type: Countries → Analysis: Citation. Full counting was selected for counting, association strength for normalization, and default settings for clustering/positioning. To illustrate the citations received by countries based on their publications, a network map was created using at least one publication and one citation criterion (threshold value). The resulting network map is given in Figure 3.

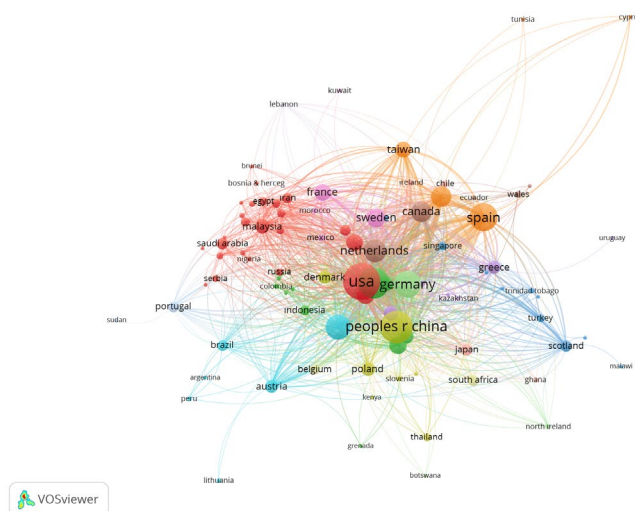


Figure 3. Citation network of countries

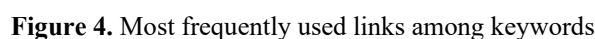
The analysis found 14 clusters, 1,052 connections, and a TLS of 6,239. When the resulting dataset was sorted from largest to smallest according to total citation counts, the top 5 countries that scientifically guided the literature and were most frequently cited were: the USA (16,439 citations), England (16,078 citations), Germany (12,692 citations), Spain (11,938 citations), and China (11,461 citations). This ranking demonstrated that these countries were global “centers of excellence” in the field of ERP and sustainability integration, and that the written theoretical/empirical frameworks formed the backbone of the global literature.

Table 3 presents the results of the citation analysis conducted on a country-by-country basis, highlighting the geographical and intellectual centers of gravity in the ERP and sustainability/ESG literature. In the global ranking based on total citation numbers, the USA was clearly in first place with 16,439 citations, rendering it the global locomotive of the literature. The USA was followed by England, Germany, Spain, and China based on the total citations.

Country	Publications	Citations	TLS
USA	291	16,439	1,027
England	202	16,078	1,041
Germany	155	12,692	911
Spain	164	11,938	873
China	226	11,461	575

An examination of this intensely global competitive network ranked Turkey 37th, with a total of 617 citations out of 2,035 articles in the dataset. Turkey as an emerging economy represented the speed at which the country's industry and academia are adapting to global trends of digital sustainability. This impact level of 617 citations indicates that Turkish researchers are beginning to connect with mainstream global literature. Unlike the theoretical maturity seen in the Western world, the literature in Turkey is still in the growth stage and operational application phase.

This analysis scanned all keywords in 2,035 articles based on their relationships with one another and illustrated the concepts discussed together worldwide in the field of ERP and sustainability (conceptual map) as well as the latest trends (time analysis). In addition to the distribution of publications by year, type, subject area, and country, examining the keyword analysis was pivotal to witness the conceptual focus of the literature. WoS output, Full Record, and Cited References, were transferred to VOSviewer as plain text. For analysis, the following commands were given: Create map → Create a map based on the bibliographic data → WoS files → Type: Author keywords → Analysis: Co-occurrence. The counting was selected as full counting and the normalization was selected as association strength. The resulting network map is given in Figure 4.



As shown in Table 4, the top five keywords ranked by frequency and connection strength were sustainability, ERP, CSR, stakeholder engagement, and environmental policy. This distribution suggested that the theoretical

legitimacy of the ERP and sustainability literature was established upon three fundamental strategic axes.

Table 4. Indicators of keywords

Keyword	Occurrences	TLS
Sustainability	501	731
ERP	208	197
CSR	188	332
Stakeholder engagement	106	228
Environmental	105	213

Note: ERP = Enterprise Resource Planning; CSR = Corporate Social Responsibility; TLS = Total Link Strength.

From the perspective of Dynamic Capabilities Theory (Teece et al., 1997), ERP are positioned as enablers that capture, process, and transform unstructured environmental data into strategic corporate capabilities in the green transformation processes of businesses. ERP is no longer an operational engine, but an intellectual building block that forms the database of corporate sustainability.

The consecutive prominence of CSR and Stakeholder Engagement concepts demonstrated the inseparable links between the field and Stakeholder Theory (Freeman, 1984) and Institutional Theory (DiMaggio & Powell, 1983). The motivation behind businesses' transformation of their ERP infrastructures towards sustainability stemmed not only from the pursuit of internal efficiency, but also from the need to respond to stakeholders' demands for transparency and to gain social legitimacy.

The sustainability umbrella was led by the fifth-ranked Environmental Policy concept. This confirmed that the development of the literature was dictated by macro-level international environmental policies, emission trading systems, and legal obligations. Businesses manage the obligations imposed by macro-level environmental policies (regulatory pressures) at the micro level with the data processing power of ERP. In summary, this resulting topological structure of nine clusters demonstrated with perfect causal link that the pressure created by macro-level environmental policies was transferred into business through corporate, CSR and Stakeholder Engagement channels. ERP was positioned at the center of the operational response to this pressure.

An analysis of the time-based network map (overlay visualization) showing the distribution of keywords over time algorithmically confirmed that 2020 represented a radical paradigm shift in the chronological course of the literature. Prior to 2020, the ecosystem was shaped around more traditional, theoretical, and static sustainability/CSR concepts, while the bright yellow nodes representing the post-2020 period revealed that the field had evolved into a completely dynamic, empirical, and technology-focused phase.

The current areas of study clustered most densely in bright yellow on the map were crystallized as: digital transformation, cloud ERP, AI, decision making, information systems, circular economy, and energy efficiency. This conceptual pattern explained the current transformation in management science literature with a perfect causal link.

- Technological driving forces (AI, cloud ERP, information systems, and digital transformation): Cloud-based architectures, replacing traditionally local ERP systems, and AI algorithms are seen as central to enterprise information systems. This integration enables the real-time processing of big data, moving sustainability from theory to operation.

- Managerial output (decision making): It is understood that this real-time and transparent data is used as leverage to optimize the strategic green decision-making processes of top management.

- Sustainability goals (circular economy and energy efficiency): It is empirically confirmed that the ultimate goal of all this digital infrastructure is not abstract carbon rhetoric but rather building measurable circular economy models within the business and achieving energy efficiency.

In conclusion, this temporal topology methodologically demonstrates that the ERP and sustainability literature has now reached a mature stage and that future research will focus entirely on the operational output of AI-powered and cloud-based digital sustainability systems.

4.5 Citation Analysis of Organizations

A citation analysis of organizations was conducted to determine the impact levels of scientific output in the ERP and sustainability/ESG literature at the organizational level and to identify global research centers. WoS output, full record, and cited references were transferred to VOSviewer as plain text. The analysis was performed using the following steps: Create map → Create a map based on the bibliographic data → WoS files → Type: Organizations → Analysis: Citation. Full counting was selected for the counting, and association strength was selected for the normalization; default settings were used for placement/clustering. To determine inter-organizational citation relationships, a network analysis was performed on 2,219 observation units that have produced at least one publication and received at least one citation (threshold value). The network analysis performed is shown in Figure 5.

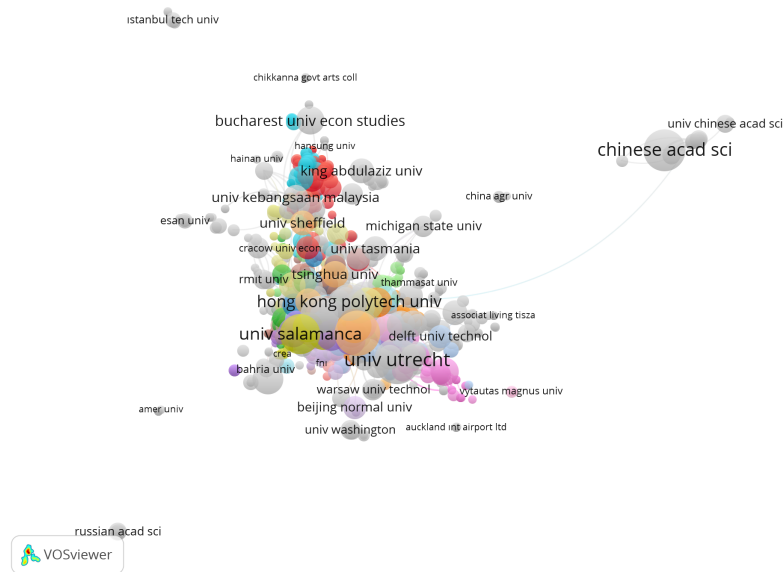


Figure 5. Citation links of institutions

This analysis showed which academic institutions offered the highest qualitative contribution (academic prestige) to the production of theoretical and practical knowledge in the field.

As shown in Table 5, the universities were ranked in descending order according to their total number of citations. The top five universities, which were the most frequently cited and intellectually prominent in the field, were identified as follows: University of Leeds (2,578), Utrecht University (2,342), University of Salamanca (2,152), Sapienza University of Rome (1,869), and University of Würzburg (1,694). The studies published by these institutions served as models of corporate excellence for researchers examining the role of ERP systems in green transformation. When the network structure was examined in general, it has been validated that the institutions with the most citations showed a complete geographical parallel with the USA, the UK, and Continental European countries that emerged as leaders in previous analyses.

Table 5. Citation analysis of institutions

Name of Institution	Publications	Citations	TLS
University of Leeds	26	2,578	163
Utrecht University	27	2,342	213
University of Salamanca	20	2,152	177
Sapienza University of Rome	6	1,869	86
University of Würzburg	5	1,694	88

Note: TLS = Total Link Strength.

When examined at the national level, Kocaeli University was seen to have the highest academic impact and citation count in the global literature among Turkish institutions. This finding indicated that the institution was a pioneer in the literature on digital sustainability and corporate systems in Türkiye. The fundamental factor behind this institutional success could be attributed to the fact that the Kocaeli region is Türkiye's largest industrial and manufacturing heartland. The intense industrial activity in the region may have facilitated the production of field-based, empirical, and high value-added studies by the university researchers on topics such as ERP systems, Industry 4.0 applications, and green supply chain management.

4.6 Author Co-Citation Analysis

Co-citation Analysis aims to reveal the connections between academic works that were cited simultaneously. While similarities were identified, the differences of studies in the same field that were not cited together could also be understood (Köseoglu et al., 2015). WoS output, full record, and cited references were transferred to VOSviewer as plain text. For analysis, the following commands were given: Create map → Create a map based on the bibliographic data → WoS files → Type: Cited authors → Analysis: Co-citation. Counting was selected as full counting, normalization as association strength, and default settings for placement/clustering. The network analysis performed is shown in Figure 6.

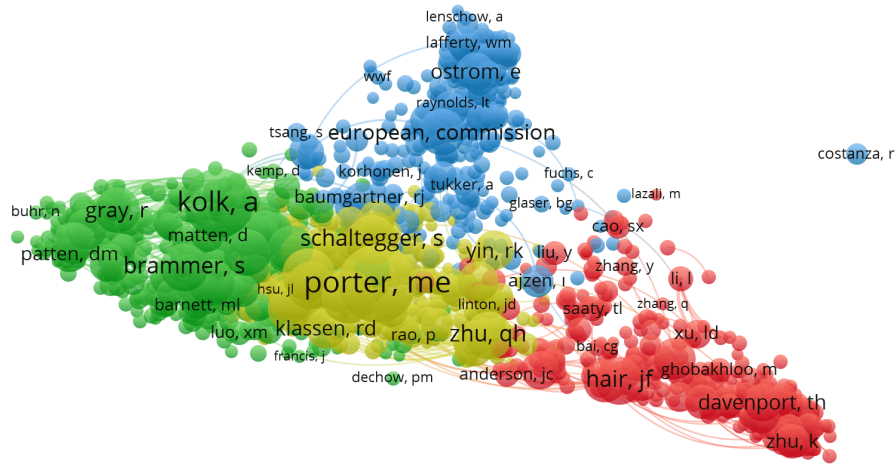


Figure 6. Collaborative citation links among authors

Based on the analysis of 3,954 units with a minimum citation count of 10 (threshold value), a total of 5 clusters, 1,580,348 connections, and a total connection power of 4,196,065 were identified. Author co-citation analysis was conducted to identify the intellectual foundation, core theories, and established scientific paradigms that had shaped the development of the ERP and sustainability literature. This analysis mapped the invisible colleges that formed the theoretical backbone of the field by directly examining the citation patterns in the references of 2,035 articles.

As shown in Table 6, the authors were ranked according to co-citation frequency. The five most frequently co-cited authors, whose works form the theoretical foundation of the field, were identified as follows: Porter, M. (390), Kolk, A. (303), Sharma, S. (186), Bansal, P. (179) and Hart, S. (147). The works of these scientists were of a “reference source/constitution” nature in the integration of ERP systems with environmental performance and the construction of corporate sustainability theories. The clusters that emerged in the map systematically revealed which theoretical schools (e.g., Enterprise Theory, Resource-Based View, or Stakeholder Theory) the researchers relied on to construct the relationship between ERP and sustainability.

Table 6. Indicators of author co-citation analysis

Author	Citations	TLS
Porter, M.	390	8,153
Kolk, A.	303	7,067
Sharma, S.	186	4,929
Bansal, P.	179	4,693
Hart, S.	147	3,736

Note: TLS = Total Link Strength.

4.7 Bibliographic Coupling of Documents

To perform bibliographic coupling analysis in VOSviewer, the conditions of two independent sources citing the same work was required. A criterion of receiving at least one citation was applied accordingly (Ağgöl et al., 2024).

WoS output, Full Record, and Cited References were transferred to VOSviewer as plain text. For analysis, the command Create map → Create a map based on the bibliographic data → WoS files → Type: Documents → Analysis: Bibliographic coupling was given. The counting was selected as full counting, the normalization as association strength, and the default settings of the program for placement and clustering. The network analysis performed is shown in Figure 7.

Bibliographic coupling of documents was performed to identify actors in the ERP and sustainability literature who drew from similar bibliographies, share common intellectual origins, and shape current research fronts. This analysis considered both the citations parameter, which represented the absolute visibility of authors in the global literature, and the TLS parameter, which indicated the intensity of bibliographic similarity links with other researchers in the dataset.

As shown in Table 7, the top five authors worldwide, ranked by total citation count, were: Dempsey et al. (2011), Evans et al. (2017), Le Blanc (2015), Schaltegger (2011), and Young et al. (2010). These researchers are positioned as “paradigmatic pioneers” who established the theoretical foundation of the intersection of ERP and sustainability, and whose works were absolutely referenced in every new study in the field.

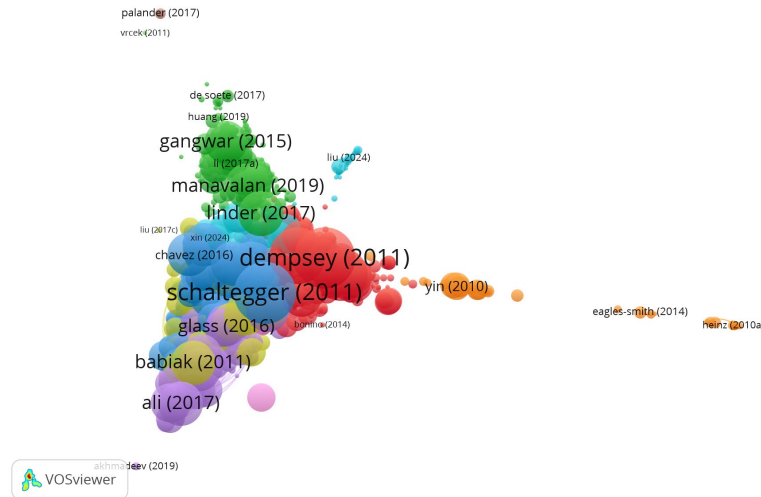


Figure 7. Network illustrating the bibliographic matching of texts

Table 7. Bibliographic matching indicators of texts (by citation)

Article	Citations	TLS
Schaltegger (2011)	1,222	360
Dempsey et al. (2011)	1,182	137
Young et al. (2010)	1,027	117
Le Blanc (2015)	900	100
Evans et al. (2017)	811	455

Note: TLS = Total Link Strength.

As shown in Table 8, the five authors with the highest TLS values, indicating the strongest bibliographic similarity with other authors, were: Dögl & Behnam (2015), Fifka (2013), Gallego-Álvarez et al. (2014), Oberhofer & Dieplinger (2014), and Wolf (2014). When evaluated within the context of dynamic capabilities and organizational theories, these authors with high TLS values served as the current “epicenters” and “intellectual cement” of the literature. The high structural similarity in the bibliographies of these authors testified the existence of a homogeneous research clique that addressed ERP-based digital sustainability transformation with the most current and holistic literature base.

Table 8. Bibliographic matching indicators of texts (based on TLS)

Article	Citations	TLS
Fifka (2013)	377	1,854
Wolf (2014)	109	1,688
Oberhofer & Dieplinger (2014)	36	1,645
Dögl & Behnam (2015)	85	1,614
Gallego-Álvarez et al. (2014)	107	1,554

Note: TLS = Total Link Strength.

5. Discussion

This study provided a literature review for researchers working on the intersection of ERP and sustainability, while also offering ERP system administrators a vision of the global literature on how to integrate digital transformation processes with sustainability goals. A total of 2,035 articles were obtained from the WoS database using bibliometric network analysis techniques to decipher the map, semantic structure, and temporal evolution of the global literature at the intersection of ERP and enterprise sustainability. Bibliometrics is a field of study that systematically maps the interactions and contributions of authors, academic works, countries, and institutions within a discipline by examining quantitative scientific studies published in a specific time. This method, in addition to visualize the evolutionary process as well as co-author and citation networks in the literature, equips researchers with a comprehensive perspective to identify current trends in the literature. It could track changes in research topics over time and identify themes that are rising or losing importance (Aksay & Ağtaş, 2025).

Subsequent to data cleaning, keyword network analysis discovered 122 connections, 2,485 TLS, and 9 thematic clusters. This observation empirically argued that the socio-technical ecosystem under study did not exhibit a

random distribution but was rather based on a highly integrated and mature epistemological foundation. In the keyword network analysis, the first large cluster belonged to the concept of sustainability. This was followed by the ERP concept cluster. This structure, from the perspective of Dynamic Capabilities Theory (Teece et al., 1997), suggested that modern ERP systems were no longer seen as merely static processing engines that minimized operational costs; rather, they were the main factor in transforming complex environmental data into strategic corporate capabilities in green transformation processes. Dempsey et al. (2011) contended that current discussions on sustainability took a holistic approach that encompassed economic and social dimensions, instead of limiting the concept to merely environmental issues. ERP systems are considered more than just operational tools; they are a strategic support system that enables businesses to increase resilience and efficiency in the long term, as well as meeting regulatory requirements (Pugna & Boldeanu, 2025). Similarly, Wolf (2014) advocated that sustainable supply chain management could strengthen a company's reputation as a "good corporate citizen". Therefore, he presented arguments against the notion that external pressure from stakeholders was the sole driving force behind sustainable supply chain management practices. The current findings implied that stakeholders' pressure and sustainable supply chain management practices significantly contributed to the sustainability performance of organizations. In this light, sustainable supply chain management should be considered by supply chain managers as a variable that provides strategic value, going beyond risk mitigation. In the keyword network analysis, cluster examination reported that CSR ranked third, Stakeholder Engagement ranked fourth, and Environmental Policy ranked fifth. When this result was assessed within the contexts of Corporate Theory (DiMaggio & Powell, 1983) and Stakeholder Theory (Freeman, 1984), it could be interpreted that macro-level pressures and stakeholders' demands for transparency were forcing businesses towards digital integration. ERP functions as an intellectual building block, facilitating the flow of data between macro-level policies and the micro-level pursuit of corporate legitimacy. According to Fatima & Elbanna (2023), as CSR became gradually prevalent and important in organizational strategies, both academic circles and the business world would benefit from implementing these practices. S-ERP systems offer holistic solutions encompassing the environmental, social, and economic dimensions of corporate sustainability. These systems rewarded with significantly strategic advantages such as improved resource efficiency, enhanced compliance with regulations, and increased stakeholder engagement (Alzahmi et al., 2025). ERP applications help protect corporate reputation from potential financial penalties by optimizing compliance with environmental regulations and international sustainability standards, as well as automating reporting processes. Furthermore, these systems allow a more rigorous evaluation of suppliers who meet sustainability criteria by analyzing resource consumption trends (Barna, 2024).

Bibliographic matching analysis of the texts revealed that the literature was built upon two fundamental intellectual pillars with chronological and methodological consistency. Schaltegger (2011) and Dempsey et al. (2011), having gained the highest number of citations, have become the leading authors in the literature. On the other hand, Fifka (2013) and Wolf (2014), with the highest TLS values, have served as methodological linkers of the literature.

6. Conclusions

When evaluated in general terms, it is worth noting that ERP systems have become more than just tools for improving efficiency in operational processes; they have indeed become fundamental components of corporate governance and risk management. ERP software integrates corporate data from a holistic perspective, supporting the concepts of transparency and accountability as well as indispensable elements of governance with concrete data. Current trends in the literature proposed that advanced ERP systems integrated ESG data into a digital platform, to offer auditable and error-free sustainability reporting capabilities. Furthermore, the "Green ERP" functions integrated into the system enable real-time monitoring of critical indicators such as carbon footprint, use of resources, and legal compliance, thus minimizing companies' environmental and financial risks and reinforcing their risk management strategies. In conclusion, the interaction between ERP and sustainability should be considered not merely as a technical integration process, but rather a strategic managerial competency that minimizes corporate risks and guarantees transparent governance at specific time.

Like any empirical studies, this research has certain limitations. The primary limitation is that the analysis data was drawn solely from the WoS Core Collection database and only peer-reviewed articles were included. The keywords used in the data collection process were customized based on Boolean logic to ensure direct relevance to the topic. This coding logic might have excluded studies in the literature that dealt with similar topics using different concepts. Additionally, the selection of English only publications in the form of articles and reviews is another limitation. Furthermore, the exclusion of medical, psychological, and natural science fields from the research scope constituted another constraint. Yet, since the data was retrieved on November 26, 2025, subsequent updates to WoS might lead to limited fluctuations in the total number of publications and citations.

Future studies should conduct comparative meta-analyses by including data from Scopus and Google Scholar to increase the depth of the literature. The output presented by the time-based (overlay visualization) network map showed that 2020 marked a radical paradigm shift in the literature. Discussions before 2020 gave way to

technological trends represented by bright yellow nodes. Accordingly, future research could examine the transformative role of AI algorithms processing big data in real-time sustainability decision-making processes in systems where cloud ERP and AI integration is present. Furthermore, by focusing on circular economy and energy efficiency, researchers could empirically investigate how enterprise information systems could be transformed into measurable circular economy output and real-time energy efficiency by optimizing internal waste management.

Author Contributions

Conceptualization, G.O. and H.M.K.; methodology, G.O.; software, G.O. and H.M.K.; validation, G.O. and H.M.K.; formal analysis, G.O.; investigation, G.O.; resources, H.M.K.; data curation, H.M.K.; writing—original draft preparation, G.O.; writing—review and editing, G.O.; visualization, G.O. All authors have read and agreed to the published version of the manuscript.

Data Availability.

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- Agarwal, P. & Gupta, A. (2024). Harnessing the power of enterprise resource planning (ERP) and customer relationship management (CRM) systems for sustainable business practices. *Int. J. Comput. Trends Technol.*, 72(4), 102–110. <https://doi.org/10.14445/22312803/%20IJCTT-V72I4P113>.
- Ağgül, E., Yalçın, P., & Yalçın, S. A. (2024). Metaverse kavramı ve öne çıkan terimler üzerine bibliyometrik bir analiz. *IBAD*, 16, 126–153. <https://doi.org/10.21733/ibad.1478575>.
- Aksay, M. & Ağtaş, E. (2025). Kurumsal yönetim ve finansal performans: 2005–2025 yılları arası WoS makalelerinin bibliyometrik analizi. *Ulus. Yön. Akad. Derg.*, 8(3), 802–819. <https://doi.org/10.33712/mana.1777175>.
- Alzahmi, W., Al-Assaf, K., Al Sheikh, R., & Bahroun, Z. (2024). Sustainable ERP systems: Current trends, challenges, and future directions. In *2024 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)* (pp. 1–8). Sharjah, United Arab Emirates. <https://doi.org/10.1109/ICTMOD63116.2024.10878242>.
- Alzahmi, W., Al-Assaf, K., Alshaikh, R., & Bahroun, Z. (2025). Towards sustainable ERP systems: Emerging trends, challenges, and future pathways. *Manag. Syst. Prod. Eng.*, 33(1), 24–38. <https://doi.org/10.2478/mspe-2025-0003>.
- Anaya, L., Flak, L., & Abushakra, A. (2023). Realizing sustainable value from ERP systems implementation. *Sustainability*, 15(7), 5783. <https://doi.org/10.3390/su15075783>.
- Anjaria, K. (2025). *Sustainable Enterprise Resource Planning (S-ERP) for Industry 4.0: A Secure and Ethical Deployment Approach*. Springer Nature. <https://doi.org/10.1007/978-981-96-7734-4>.
- Barna, L. E. L. (2024). ERP systems and their role in corporate sustainability management. In *Development Through Research and Innovation* (pp. 379–384). <https://doi.org/10.53486/dri2024.43>.
- Bhagat, S. K. (2025). Ethical integration of AI in public sector ERP: Balancing innovation with accountability. *Int. J. Inf. Technol. Manag. Inf. Syst.*, 16(2), 254–268. https://doi.org/10.34218/IJITMIS_16_02_018.
- Chofreh, A. G., Goni, F. A., Klemeš, J. J., Malik, M. N., & Khan, H. H. (2020). Development of guidelines for the implementation of sustainable enterprise resource planning systems. *J. Clean. Prod.*, 244, 118655. <https://doi.org/10.1016/j.jclepro.2019.118655>.
- Christmann, A. S., Crome, C., Graf-Drasch, V., Oberländer, A. M., & Schmidt, L. (2024). The twin transformation butterfly: Capabilities for an integrated digital and sustainability transformation. *Bus. Inf. Syst. Eng.*, 66(4), 489–505. <https://doi.org/10.1007/s12599-023-00847-2>.
- Clark, G. L., Feiner, A., & Viehs, M. (2015). *From the stockholder to the stakeholder: How sustainability can drive financial outperformance*. SSRN. <https://doi.org/10.2139/ssrn.2508281>.
- Dempsey, N., Bramley, G., Power, S., & Brown, C. (2011). The social dimension of sustainable development: Defining urban social sustainability. *Sustain. Dev.* 19(5), 289–300. <https://doi.org/10.1002/sd.417>.
- DiMaggio, P. J. & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *Am. Sociol. Rev.*, 48(2), 147–160. <https://doi.org/10.2307/2095101>.
- Dögl, C. & Behnam, M. (2015). Environmentally sustainable development through stakeholder engagement in developed and emerging countries. *Bus. Strat. Environ.*, 24(6), 583–600. <https://doi.org/10.1002/bse.1839>.

- Donthu, N., Kumar, S., Mukherjee, D., Pandey, N., & Lim, W. M. (2021). How to conduct a bibliometric analysis: An overview and guidelines. *J. Bus. Res.*, 133, 285–296. <https://doi.org/10.1016/j.jbusres.2021.04.070>.
- Ebirim, G. U., Unigwe, I. F., Asuzu, O. F., Odonkor, B., Oshioste, E. E., & Okoli, U. I. (2024). A critical review of ERP systems implementation in multinational corporations: Trends, challenges, and future directions. *Int. J. Manag. Entrep. Res.*, 6(2), 281–295. <https://doi.org/10.51594/ijmer.v6i2.770>.
- Ekins, P. (1999). *Economic Growth and Environmental Sustainability: The Prospects for Green Growth*. Routledge.
- Evans, S., Vladimirova, D., Holgado, M., Van Fossen, K., Yang, M., Silva, E. A., & Barlow, C. Y. (2017). Business model innovation for sustainability: Towards a unified perspective for creation of sustainable business models. *Bus. Strat. Environ.*, 26(5), 597–608. <https://doi.org/10.1002/bse.1939>.
- Fatima, T. & Elbanna, S. (2023). Corporate social responsibility (CSR) implementation: A review and a research agenda towards an integrative framework. *J. Bus. Ethics*, 183(1), 105–121. <https://doi.org/10.1007/s10551-022-05047-8>.
- Fifka, M. S. (2013). Corporate responsibility reporting and its determinants in comparative perspective—A review of the empirical literature and a meta-analysis. *Bus. Strategy Environ.*, 22(1), 1–35. <https://doi.org/10.1002/bse.729>.
- Freeman, R. E. (1984). *Strategic Management: A Stakeholder Approach*. Pitman.
- Gallego-Álvarez, I., García-Sánchez, I. M., & da Silva Vieira, C. (2014). Climate change and financial performance in times of crisis. *Bus. Strat. Environ.*, 23(6), 361–374. <https://doi.org/10.1002/bse.1786>.
- Hasan, M. S., Ebrahim, Z., Mahmood, W. W., & Ab Rahman, M. N. (2017). Sustainable-ERP system: A preliminary study on sustainability indicators. *J. Adv. Manuf. Technol.*, 11(1), 61–74.
- Koçyiğit, S. Ç., Temelli, F., & Baskan, T. D. (2023). Sürdürülebilirlik muhasebesi konusunda yayınlanan makalelerin bibliyometrik analizi: Web of Science örneği. *Ömer Halisdemir Üniv. İktisadi İdari Bilim. Fak. Derg.*, 16(1), 241–264. <https://doi.org/10.25287/ohuibf.1188162>.
- Köseoğlu, M. A., Sehitoglu, Y., & Craft, J. (2015). Academic foundations of hospitality management research with an emerging country focus: A citation and co-citation analysis. *Int. J. Hosp. Manag.*, 45, 130–144. <https://doi.org/10.1016/j.ijhm.2014.12.004>.
- Landauer, T. K., Foltz, P. W., & Laham, D. (1998). An introduction to latent semantic analysis. *Discourse Process.*, 25(2–3), 259–284. <https://doi.org/10.1080/01638539809545028>.
- Le Blanc, D. (2015). Towards integration at last? The sustainable development goals as a network of targets. *Sustain. Dev.*, 23(3), 176–187. <https://doi.org/10.1002/sd.1582>.
- Lin, H. Y., Hsu, P. Y., & Ting, P. H. (2006). ERP systems success: An integration of IS success model and balanced scorecard. *J. Res. Pract. Inf. Technol.*, 38(3), 215–228.
- Lotfy, M. A. M. B. (2015). *Sustainability of enterprise resource planning (ERP) benefits postimplementation: An individual user perspective* [Doctoral thesis, Walden University]. <https://scholarworks.waldenu.edu/dissertations/1316/>
- Oberhofer, P. & Dieplinger, M. (2014). Sustainability in the transport and logistics sector: Lacking environmental measures. *Bus. Strat. Environ.*, 23(4), 236–253. <https://doi.org/10.1002/bse.1769>.
- Odenwald, T. & Berg, C. (2014). A new perspective on enterprise resource management. *MIT Sloan Manag. Rev.*, 56(1), 12.
- Özari, Ç. & Can, E. N. (2023). Sürdürülebilirlik muhasebesi ve karbon muhasebesi çalışmalarının bibliyometrik analizi. *Denetişim*, 27, 55–71. <https://doi.org/10.58348/denetisim.1177406>.
- Pérez Estébanez, R. (2024). An approach to sustainable enterprise resource planning system implementation in small-and medium-sized enterprises. *Adm. Sci.*, 14(5), 91. <https://doi.org/10.3390/admsci14050091>.
- Praharaj, A. & Chhatoi, B. P. (2025). Enterprise resource planning (ERP) in business: A contextual snapshot. *Glob. Knowl. Mem. Commun.* <https://doi.org/10.1108/GKMC-07-2024-0453>.
- Pugna, I. B. & Boldeanu, D. M. (2025). The role of ERP systems in driving corporate sustainability. In *Proceedings of the 19th International Conference on Business Excellence 2025* (pp. 402–412). Sciendo, Bucharest, Romania. <https://doi.org/10.2478/picbe-2025-0033>.
- Schaltegger, S. (2011). Sustainability as a driver for corporate economic success. Consequences for the development of sustainability management control. *Soc. Econ.*, 33(1), 15–28. <https://doi.org/10.1556/socec.33.2011.1.4>.
- Teece, D. J., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strateg. Manag. J.*, 18(7), 509–533. [https://doi.org/10.1002/\(SICI\)1097-0266\(199708\)18:7%3C509::AID-SMJ882%3E3.0.CO;2-Z](https://doi.org/10.1002/(SICI)1097-0266(199708)18:7%3C509::AID-SMJ882%3E3.0.CO;2-Z).
- Tilt, C. A. (2016). Corporate social responsibility research: The importance of context. *Int. J. Corp. Soc. Responsib.*, 1(1), 2. <https://doi.org/10.1186/s40991-016-0003-7>.
- United Nations. (2015). *The 17 goals*. <https://sdgs.un.org/goals>
- Van Eck, N. J. & Waltman, L. (2010). Software survey: VOSviewer, a computer program for bibliometric mapping. *Scientometrics*, 84(2), 523–538. <https://doi.org/10.1007/s11192-009-0146-3>.

- Wang, C., Zhang, Y., Liu, C., & Yao, S. (2025). *The impact of artificial intelligence on enterprise ESG performance: The moderating role of digital transformation?* SSRN. <https://doi.org/10.2139/ssrn.5135864>.
- Wolf, J. (2014). The relationship between sustainable supply chain management, stakeholder pressure and corporate sustainability performance. *J. Bus. Ethics*, 119(3), 317–328. <https://doi.org/10.1007/s10551-012-1603-0>.
- Young, W., Hwang, K., McDonald, S., & Oates, C. J. (2010). Sustainable consumption: Green consumer behaviour when purchasing products. *Sustain. Dev.*, 18(1), 20–31. <https://doi.org/10.1002/sd.394>.
- Yüzgenç, I. U. & Aydemir, E. (2023). Sustainable ERP systems: A green perspective. In *International Conference on Pioneer and Innovative Studies* (Vol. 1, pp. 533–538). Konya, Turkey. <https://pdfs.semanticscholar.org/faca/7ec0f027cd6fe1963713e70e82ba678f5f45.pdf>